

WHEN THE SOUTH AFRICAN STATISTICAL ASSOCIATION WAS FOUNDED*

H.S. Steyn

Department of National Education

Mr President, as a foundation member it is a real pleasure for me to comply with the request of the South African Statistical Association to address you on this occasion, the twenty-fifth anniversary of the association which was founded on 28 October 1953.

The first revival in mathematical statistics in our country came relatively early in the history of modern mathematical statistical theory. Round about the nineteen thirties four South African students took their doctors' degrees in probability theory or mathematical statistics in Great Britain. They were J.H. Müller, S.J. Pretorius, J.M. le Roux and H.T. Gonin — all graduates from the University of Stellenbosch. The research work done by S.J. Pretorius under the guidance of Karl Pearson and published in *Biometrika* (1930) is perhaps best known and is still quoted today in well-known textbooks** Pretorius was well-known first as professor in Statistics at Stellenbosch and from 1954 till his retirement as Registrar of that University. We think of him today with appreciation for his pioneering work and wish him well, especially as we understand that he is in less good health.

Le Roux, for many years, was professor in Applied Mathematics at the University of Stellenbosch while Müller played some part in the industrial development of our country.

Some of you no doubt remember the late H.T. Gonin who, after his return to South Africa from Edinburgh proceeded to study theology and, although doing missionary work, still remained interested in statistical theory and sporadically wrote articles for local and overseas journals (e.g. *Biometrika*, 1961).

At most of our universities up to 1950 it was the mathematicians with a limited interest in statistical theory and probability who did valuable work to promote the interest in the Theory of Statistics. B. de Loor who, in the twenties, obtained his doctorate in algebra under the guidance of the well-known Dutch mathematician and philosopher Brouwer, came back to the University of Pretoria where he became interested in probability and statistics and laid the foundations for the later strong Department of Statistics. Quite a number of statisticians present today were students of De Loor. Other mathematicians that may be mentioned in this connection is D. Burnett of Rhodes University, A.P. Malan of the University of the Orange Free State, who studied under R.A. Fisher and under whose guidance it was my privilege to start my career, D.J. van Rooy of the Potchefstroom University for Christian Higher Education and P. Stein of the

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University of Natal. While mentioning the University of Natal it is necessary also to mention the pioneering work of A.R. Saunders in Biometry.

It is not possible to talk about statistical theory and its development in South Africa prior to 1950 without mentioning the impact of the publication of the two volumes by M.G. Kendall on the Advanced Theory of Statistics in 1942 and 1946, and I think this is true for most countries in the world. These two volumes together with the mimeograph textbook by S.S. Wilks which appeared about the same time, brought modern statistical theory to our doorsteps.

The above have been mentioned as background only. I hope that I have succeeded in making it clear that, while considerable attention was being given to statistics both from the mathematical as well as from the descriptive side, no real contribution to the theory of statistics was made in South Africa before the nineteen forties. The breakthrough came round about 1941 when a lecturer from the mathematics department of the University of the Witwatersrand, as an internee at Hald in Denmark with enough time at his disposal, started to spin a small coin ten thousand times and to record the results. This experiment was the starting point of experiments leading to J.E. Kerrich's book: "An Experimental Introduction to the Theory of Probability", and by doing so, John Kerrich succeeded in writing a book that has laid the foundation for statistical orderly thinking, for many generations of students taking statistics for a variety of reasons at the University of the Witwatersrand and at other South African Universities.

The research work by H.S. Sichel and A.G. Arbous at the beginning of the fifties on the statistical theory of absenteeism made a valuable contribution to statistical theory. Their publication in *Biometrika* (1954) is still widely referred to. Apart from what was being done at the universities in the early fifties, there also came into operation a number of units doing applied research in mathematical statistics. I mention only a few from memory, knowing that I may leave out some that should have been mentioned: I refer here to the unit at the National Institute for Personnel Research, at the South African Bureau of Standards, at the Chamber of Mines Laboratories, at Modderfontein, at Onderstepoort, at Iscor and at the Department of Agriculture. As a result of these activities a small but regular stream of research reports started to appear in local and overseas scientific journals and the statisticians, although limited in number, felt the need for closer contact.

A suitable occasion for the formation of a South African Association came in October 1953 when Gertrude M. Cox from the University of North Carolina paid a visit to our country and at 8 p.m. on 28 October 1953 at the invitation of Kerrich a number of us gathered together at the University of Witwatersrand. Present were: Miss G.M. Cox, B. de Loor, J.E. Kerrich, H.S. Sichel, A.G. Arbous, J.S. Maritz, G. Abraham, P. Cristensen, A.M. Adelstein, D.G. Krige, P. Armsen, W. Lutz and myself. It was quite an easy task to elect the office bearers namely: Honorary President: Miss Cox, President: B. De Loor, Vice-President: J.E. Kerrich, Secretary/Treasurer: J.S. Maritz and as further members of the executive committee, H.S. Sichel, A.G. Arbous and myself.

Our first secretary/treasurer, J.S. Maritz, then of the Chamber of Mines Laboratories and now the well-known J.S. Maritz of Monash University and previously of Melbourne University in Australia, was a very competent officer and I am sure that the sound progress made by the South African Statistical Association during its first four years, can

to a large extent be contributed to his efficiency.

The first meeting of the executive committee was held on 12 January 1954 and the first public meeting of the association took place on 18 May 1954 at the University of the Witwatersrand where the inaugural address was delivered by the president, B. De Loor. F.G. Hill took the chair for the occasion and the vote of thanks was proposed by J.M. Hyslop — still well-known names to all of us.

The opening paragraph of De Loor's address (1954) reads: "A Comparatively new branch of scientific method has come into its own, and is permeating all phases of research, industry and organization. This new branch, which is the scientific expression of logical induction, is the statistical method. Its development has been phenomenal, especially during the last two decades. To express it in terms of modern physics, a chain reaction has been set up and is making its influence felt in nearly every sphere of human activity.

Here are a few examples:—

- (i) The theory of nuclear physics is today built on the theory of probability. The deterministic standpoint of the last century has given way to the statistical standpoint with its indeterminism of the individual in the microcosm and its determinism of the average in the macrocosm.
- (ii) In agricultural research the design of experiments has been revolutionized and new ground has been broken by the impact of statistical thought on genetics, plant-breeding and allied sciences."

De Loor proceeded to give many more examples covering the medical, social and economic sciences and gave a short account on the history of Statistics from the time of the Roman Empire via Pascal, Fermat, Bernoulli, Galton and Pearson. He compared a mathematical number with a statistical number — "A mathematical number is a point. A statistical number is a point with a fringe round it". The inaugural lecture was indeed a memorable occasion.

It is interesting to note the titles of some of the papers delivered to the association during the first few years: H.L.F. Snyman (of Iscor): *Praktiese toepassings van Statistiese Kwaliteitsbeheer*. J.E. Kerrich: *Confidence limits associated with a straight line fitted by least squares*. W. Lutz (now of Edinburgh University): *Some aspects of Cluster Sampling as applied to Social Surveys*. H.M. Stoker (Secretary of Department of Statistics): "*'n Komende Ondersoek in verband met Familiebegrotings*". H.S. Sichel: *A lot from a little*. T. de Jongh (now president of the Reserve Bank): *Verbetering van die Unie se Betalings-balansskattings*.

Maybe we should think again about our present interests. We tried in the early years to get a sound balance between mathematical theory of statistics and a variety of fields of application and I am sure that this effort was a success.

The annual subscription was fixed at 10/- which was about the same proportion of my salary as a senior lecturer at that time as is the present R5 compared to the current salary of a senior lecturer. In 1957 George Abraham took over as secretary/treasurer and did a marvellous job selling statistics to industry and the public. Unfortunately George (still at C.S.I.R.) decided not to make statistics his full career. After George Abraham, W. Lutz, then lecturer at Wits, became secretary/treasurer for one year, followed by R. Wal-lach of Iscor who remained till 1966. De Loor remained president of the association for

three years when Kerrich took over for two years before handing over in turn to Sichel. At this stage the constitution was changed to provide for the present procedure whereby annually a vice-president is elected who automatically becomes the new president for the following year.

Due to the inspiration and motivation by the persons whose names I have mentioned above and of their colleagues whose names I could not all mention today, the development of statistical activities in this country and of our association were such that after twenty five years we can look back on a sound basis and on our own journal, already well-known amongst statisticians the world over. This is indeed an achievement to be proud of.

May the future be as prosperous.

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